across the whole universe: once you find the particle here, the thinking goes, the probability of its being found *anywhere else* immediately drops to zero, and this is reflected in an immediate collapse of the probability wave.

In the Aspect experiment, when the left-moving photon's spin is measured and is found, say, to be clockwise about some axis, this collapses its probability wave throughout all of space, instantaneously setting the counterclockwise part to zero. Since this collapse happens everywhere, it happens also at the location of the right-moving photon. And, it turns out, this affects the counterclockwise part of the right-moving photon's probability wave, causing it to collapse to zero too. Thus, no matter how far away the right-moving photon is from the left-moving photon, its probability wave is instantaneously affected by the change in the left-moving photon's probability wave, ensuring that it has the same spin as the left-moving photon along the chosen axis. In standard quantum mechanics, then, it is this instantaneous change in probability waves that is responsible for the faster-than-light influence.

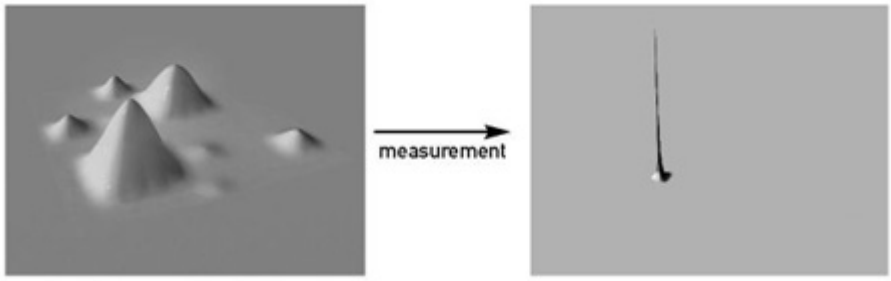


Figure 4.7 When a particle is observed at some location, the probability of finding it at any other location drops to zero, while its probability surges to 100 percent at the location where it is observed.

The mathematics of quantum mechanics makes this qualitative discussion precise. And, indeed, the long-range influences arising from collapsing probability waves change the prediction of how often Aspect's left and right detectors (when their axes are randomly and independently chosen) should find the same result. A mathematical calculation is required to get the exact answer (see notes section[18] if you're interested), but when the math is done, it predicts that the detectors should agree *precisely*50 percent of the time (rather than predicting agreement more than 50 percent of the time—the result, as